Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

[12AE71]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER–2015

AEROSPACE CONTROL SYSTEMS

Time:3 Hours **Maximum Marks:70**

Instructions:

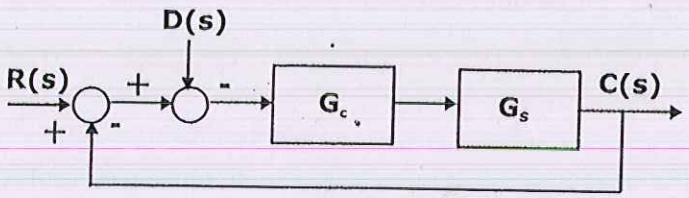
1. Answer any **FIVE** full Questions.
2. Assume suitable missing data (if any)

1. Using the partial fraction expansion approach for the transfer function given by 14

$$\frac{C(s)}{R(S)} = \frac{10}{(T_1S+1)(T_2S+1)}$$

Determine the value of c(t) at t=1 when $T_1 = T_2 = 1$ and R(s)=1/s

2. Derive the relationship between the output C(s) and the inputs R(s) and D(s) for the system shown in the block diagram below. 14



Calculate the steady value if c(t) when $G_c = K_p, G_s = \frac{5}{5s+1}$ and $R(s) = 2D(s) = \frac{1}{s}$.

3. For the system block diagram shown in problem 2, let D(s)=0, the transfer function. $G_s = \frac{5}{s^2+2s+3}$ 14

And $G_c=K_p$. What is the value of the proportional gain K_p that will give the absolute value of the steady- state error to be 0.1 when R(s)=1/s

4. Determine if all the roots of the polynomial 14

$$s^4 + 4s^3 - s^2 - 16s - 12 = 0$$

have negative real part using Routh's criterion. If not, how many roots have positive real parts?

5. Plot the root locus of the system with the open loop transfer function, 14

$$G_{(s)} = \frac{K(s+2)}{s^3+3s^2+2s}.$$

Indicate the location of the break-away point approximately on the root locus plot. It is not necessary to plot the root locus accurately

SCHOOL OF ENGINEERING & TECHNOLOGY JAIN UNIVERSITY-BANGALORE LIBRARY DATE

6. **a.** Plot the frequency response plot and the system with transfer function. 10

$$G_{(s)} = \frac{10(s+2)}{2s^3+6s^2+4s}$$

7. **a.** Determine if the system defined in the state-space form by the equation. 10

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -8 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Is controllable and observable?

8. Consider a first-order system defined by the 10

$$\frac{dx}{dt} = 2x + u$$

If the control u is selected as u=-10x, determine the eigenvalue and the closed-loop system. Compare it with the root and the characteristic equation of the closed-loop polynomial of the system with the open-loop transfer function.

$$G_c G_s = \frac{10}{s-2}$$

Table of Laplace Transforms

1(t)	$\frac{1}{s}$
e^{-at}	$\frac{1}{s+a}$
$\frac{1}{(n+1)!} t^{n-1} e^{-at}$	$\frac{1}{(s+a)^n}$